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RESEARCH

Comparative evaluation of hyaluronic acid spray and low-level laser therapy on wound healing following gingivectomy: a randomized controlled clinical trial

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Abstract

Gingivectomy wounds heal by secondary intention, prolonging recovery and postoperative discomfort; this randomized controlled trial compared low-level laser therapy and hyaluronic acid spray for early wound healing and postoperative pain after gingivectomy. Systemically healthy patients requiring gingivectomy were randomly allocated to receive either LLLT (940 nm diode laser, 3 W, 128.4 J/cm^2, applied on days 0, 3, 5, and 7) or 0.04% high-molecular-weight HA spray (applied twice daily for 7 days). Wound epithelialization was quantified through Mira-2-Ton plaque-disclosing staining combined with digital planimetric analysis (ImageJ) at baseline and on postoperative days 3, 7, and 14. Secondary outcomes included gingival index (GI), full-mouth bleeding score (FMBS), probing pocket depth (PPD), and visual analogue scale (VAS) pain scores. A total of 64 participants were enrolled and randomized ($n=32$ per group). Both groups exhibited significant wound area reduction over time ($p<0.001$). However, the HA spray group demonstrated significantly higher wound epithelialization percentages (percentage reduction from baseline in the non-epithelialized, disclosing-solution–stained wound surface area) than the LLLT group on days 3 ($p<0.001$; Cohen's d = 1.39), 7 ($p<0.001$; $d=1.12$), and 14 ($p=0.044$; $d=0.73$). The HA spray group also reported significantly lower VAS pain scores on days 1 and 3 ($p<0.05$). Periodontal parameters improved significantly from baseline to week 6 in both groups (all $p<0.001$), with no significant between-group differences at week 6 for GI ($p=0.999$), FMBS ($p=0.843$), or PPD ($p=0.718$). HA spray yielded significantly accelerated re-epithelialization and superior early pain reduction compared to LLLT following gingivectomy, suggesting it may serve as a practical, patient-friendly adjunctive alternative to laser biostimulation in postoperative wound management. Trial registration: ClinicalTrials.gov Number: **NCT07034625.**

Keywords Gingival overgrowth · Low level laser · Hyaluronic acid · Gingivectomy

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Introduction

Gingival overgrowth is a multifactorial condition associated with chronic inflammation, medication use, systemic and hormonal factors, and local irritants [1]. Orthodontic treatment is a well-recognized contributing factor, as fixed appliances promote plaque accumulation and facilitate bacterial colonization, frequently leading to gingival enlargement during treatment [2]. Excessive gingival enlargement may compromise esthetics and periodontal health by creating conditions favorable for further plaque retention [3]. When non-surgical Phase I periodontal therapy fails to achieve adequate tissue resolution, gingivectomy and gingivoplasty procedures are indicated to excise the excess tissue

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and restore physiological gingival architecture [4]. First described by Robicsek in 1918 and later refined throughout the twentieth century [5, 6], gingivectomy has evolved from a purely pocket-reduction technique to one that addresses both functional and esthetic indications [7]. However, since the surgical wound heals by secondary intention, the recovery process is inherently prolonged. Surface epithelialization following gingivectomy is generally completed within 5–14 days, whereas full epithelial maturation and connective tissue remodeling may require up to four and seven weeks, respectively [8, 9]. Consequently, the prolonged healing period and associated postoperative discomfort highlight the need for adjunctive approaches that can enhance early wound healing and improve patient comfort.

Among the adjunctive approaches explored in recent years, hyaluronic acid (HA) and low-level laser therapy (LLLT) have attracted particular attention due to their potential to enhance wound healing [10–13]. HA is a naturally occurring glycosaminoglycan and a major component of the extracellular matrix that contributes to the structural integrity of periodontal tissues [14–16]. Through its effects on cellular migration, angiogenesis, and inflammatory regulation, HA promotes tissue repair and wound healing [16]. In addition, HA has been reported to exhibit anti-inflammatory, analgesic, and bacteriostatic properties, supporting its use in both non-surgical and surgical periodontal therapies [12–14, 17, 18].

Low-level laser therapy, on the other hand, employs red or near-infrared light without inducing structural changes or significant thermal elevation [19, 20]. The therapeutic potential of LLLT was first demonstrated by Mester et al., who observed enhanced wound healing in experimental rat models [21]. LLLT promotes tissue repair primarily through photobiomodulation by enhancing cellular metabolism, extracellular matrix synthesis, fibroblast proliferation, and modulation of the inflammatory response [22–24]. Although several studies have reported favorable outcomes of LLLT in reducing postoperative pain and accelerating wound healing following periodontal surgery [7, 9, 19, 22], clinical outcomes remain dependent on treatment parameters, and a universally accepted standardized protocol has not yet been established [22, 25].

Although both HA and LLLT have individually been investigated as adjunctive approaches for enhancing periodontal wound healing, most available studies have evaluated these modalities separately. Direct comparative evidence between HA and LLLT following gingivectomy remains scarce, particularly with respect to early wound epithelialization and patient-reported postoperative pain [1, 11]. As a result, clinicians currently lack clear evidence regarding which adjunctive modality may provide greater postoperative healing benefits and improved patient comfort in this specific surgical context. Addressing this gap is important for guiding evidence-based selection of adjunctive therapies after gingivectomy. Therefore, this study was designed as a randomized controlled clinical trial to compare the effects of LLLT and high-molecular-weight HA spray on postoperative wound healing and patient-reported pain following gingivectomy. The primary outcome was wound epithelialization percentage assessed via digital planimetric analysis at postoperative days 3, 7, and 14. Secondary outcomes included gingival index (GI), full-mouth bleeding score (FMBS), probing pocket depth (PPD), and visual analogue scale (VAS).

Materials and methods

Sample size calculation

The primary endpoint for sample size estimation was the between-group difference in wound healing percentage after gingivectomy. Using an effect size of $d = 0.80$ derived from the day-30 difference between the test and control groups reported by Kohale et al. [7], a two-sided significance level of $\alpha = 0.05$, and a statistical power of 0.85, the minimum required sample size for an independent samples t-test was determined to be 58 participants ($n = 29$ per group). Although repeated-measures analyses were planned to evaluate changes over time, the independent samples t-test was selected for sample size estimation because the primary objective was to detect an intergroup difference in wound healing between the two treatment arms. To compensate for potential loss of efficiency in non-parametric analyses and possible dropout, the sample size was increased by approximately 10%, resulting in a final target sample size of 64 participants ($n = 32$ per group). The calculation was performed using G*Power version 3.1.9.6 (Heinrich Heine University Düsseldorf, Düsseldorf, Germany).

Participants

Participants were recruited from among volunteers referred to the Department of Periodontology, Faculty of Dentistry, Akdeniz University. To be eligible for inclusion, participants were required to be systemically healthy and to present with chronic inflammatory gingival hyperplasia affecting at least six teeth in a symmetrical distribution within the maxillary or mandibular anterior region, with a clinical indication for gingivectomy. Additionally, eligible participants were required to have no history of periodontitis with clinical attachment loss or radiographic bone loss and to be free from any current use of medications known to affect periodontal tissues or induce gingival hyperplasia, including antiepileptics, immunosuppressants, or calcium channel blockers.

Participants were excluded from the study if they presented with any systemic disease, were pregnant or breastfeeding, had a current smoking habit, had used systemic antibiotics within the preceding six months, or demonstrated inadequate oral hygiene, defined as a full-mouth plaque score (FMPS) of≥20%.

Randomization and blinding

Participants were randomly allocated into two groups using Sealed Envelope Ltd. online randomization software (www.sealedenvelope.com), which generated a simple randomization sequence with a 1:1 allocation ratio. Allocation concealment was ensured through sequentially numbered, sealed, opaque envelopes, which were opened only upon completion of the surgical procedure. The randomization sequence was generated by the statistician (blinded to participant allocation), who also prepared the sealed envelopes. Participant enrollment was performed by the operating surgeon (A.C.K.), who opened the corresponding envelope only after completion of the surgical procedure, ensuring allocation concealment at the time of tissue excision. Following the surgical intervention, participants received either a hyaluronic acid spray or LLLT application. Due to the distinct nature of the two interventions, both the surgeon and the patients were aware of the assigned intervention; however, the outcome assessor and the statistician were blinded to the group allocations. All surgical interventions were performed by a single experienced operator (A.C.K.), while outcome assessments were conducted by an independently calibrated examiner. Calibration of wound area measurements using ImageJ software was performed by the same examiner (I.O.), and intra-examiner reliability was evaluated. The intraclass correlation coefficient (ICC) was determined to be 0.91, indicating excellent measurement reliability.

Interventions

All participants underwent comprehensive professional prophylaxis, including supra- and subgingival scaling, rubber cup prophylaxis, and root planing where indicated, prior to surgery. Tissue healing was reassessed 3–4 weeks following initial therapy, and patients demonstrating adequate oral hygiene compliance were scheduled for gingivectomy.

All surgical procedures were performed by the same experienced operator (A.C.K.). Following administration of local anesthesia (4% articaine hydrochloride with 1:100,000 epinephrine), gingival pocket depths were marked using a Williams periodontal probe (Hu-Friedy Mfg. Co, Chicago, IL, USA). Excessive gingival tissue was removed by external bevel incisions using a No. 15 scalpel blade and a Kirkland knife. Residual gingival tissue fragments were removed with an Orban knife and microsurgical scissors when necessary. The gingival tissues were subsequently contoured to restore physiological gingival morphology. After surgery, all the patients were prescribed ibuprofen 400 mg orally (twice daily for five days postoperatively). In the HA group, adherence to twice-daily spray application was assessed at each follow-up visit through direct patient questioning. All participants reported consistent adherence to the prescribed protocol throughout the 7-day application period. In Group 1 (HA group; $n=32$), participants received a 0.04% high molecular weight hyaluronic acid spray (GENGIGEL® Forte Oral Spray, Ricerfarma S.r.l., Milan, Italy) with standardized application instructions. Participants were instructed to apply the spray to the surgical area twice per day for the first 7 postoperative days. In Group 2 (LLLT group; $n=32$), a 940 nm wavelength diode laser (Epic X, BIOLASE, Foothill Ranch, CA, USA) was applied to the surgical area on days 0, 3, 5, and 7 postoperatively. The laser parameters were configured as follows: 3 W power output, power density of 1.07 W/cm^2, a probe spot size of 2.8 cm^2, continuous wave mode, an application distance of 1 mm, and a contact-free stationary application technique. The laser was applied for 120 s per session; therefore, the total energy delivered per session was 360 J, with a corresponding energy density of 128.4 J/cm^2. These parameters are consistent with those reported in previous investigations evaluating the biostimulatory effects of LLLT on wound healing following gingivectomy [1, 26, 27]. The 940 nm wavelength falls within the established near-infrared photobiomodulation range (780–1100 nm) and has documented biostimulatory effects on gingival fibroblast metabolism, angiogenesis, and wound repair. The selected 3 W power setting was within the manufacturer-recommended adjustable power range for pain therapy applications (2.75–4.0 W). The photobiomodulatory mechanism operates through non-thermal photochemical pathways, with tissue temperature maintained below physiological threshold (≤36.5 °C), as established in the LLLT safety literature [20]. No tissue temperature monitoring was performed intraoperatively; however, no thermal adverse events or patient-reported heat discomfort were observed across all 128 laser sessions delivered in this study.

Primary outcomes: wound surface area assessment

At baseline and on postoperative days 3, 7, and 14, the surgical gingivectomy site was stained with an erythrosine-based, gluten-free plaque-disclosing agent, Mira-2-Ton® (Hager & Werken GmbH & Co. KG, Duisburg, Germany), to identify regions where the gingival epithelium was absent, damaged, or inadequately keratinized, thus distinguishing these areas from healthy keratinized gingiva [31]. Standardized clinical photographs were obtained at each evaluation time point by

the same photographer using the same digital camera system (Canon EOS 70D DSLR camera, Canon Inc., Tokyo, Japan). Patients were seated in an upright position in the dental chair, and intraoral retractors were used to provide consistent exposure of the gingivectomy area. The photographs were taken under identical operatory lighting conditions, with the dental chair light positioned in the same orientation for all images. A ruler was used to standardize the camera-to-surgical site distance at 15 cm, and the camera was positioned perpendicular to the gingival surface as much as possible to maintain a consistent shooting angulation. The same magnification, focal length, and camera settings were maintained throughout all time points. Images were captured using a standardized intraoral photography protocol, including stable hand positioning, fixed framing of the surgical area, and consistent focus on the wound surface, in order to minimize variability in image acquisition across different postoperative evaluations. Wound surface area was quantified using digital planimetric analysis with ImageJ software (National Institutes of Health, Bethesda, MD, USA). Representative clinical photographs obtained at baseline and on postoperative days 3, 7, and 14 are presented for the HA spray and LLLT groups in Figs. 1 and 2, respectively, to illustrate the standardized photographic assessment protocol.

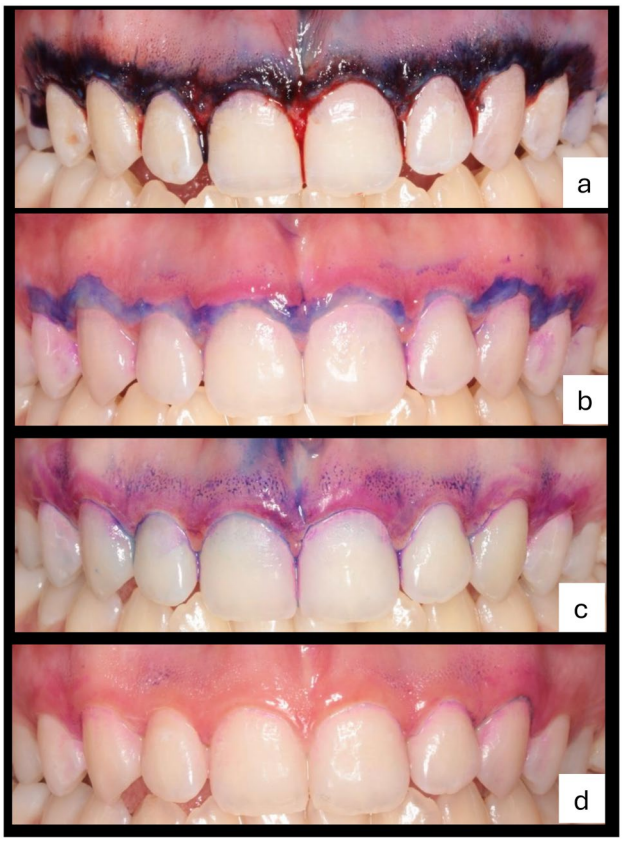


Fig. 1 Case example belonging to the HA spray group. **a**. Baseline **b**. Day 3 **c**. Day 7 **d**. Day 14

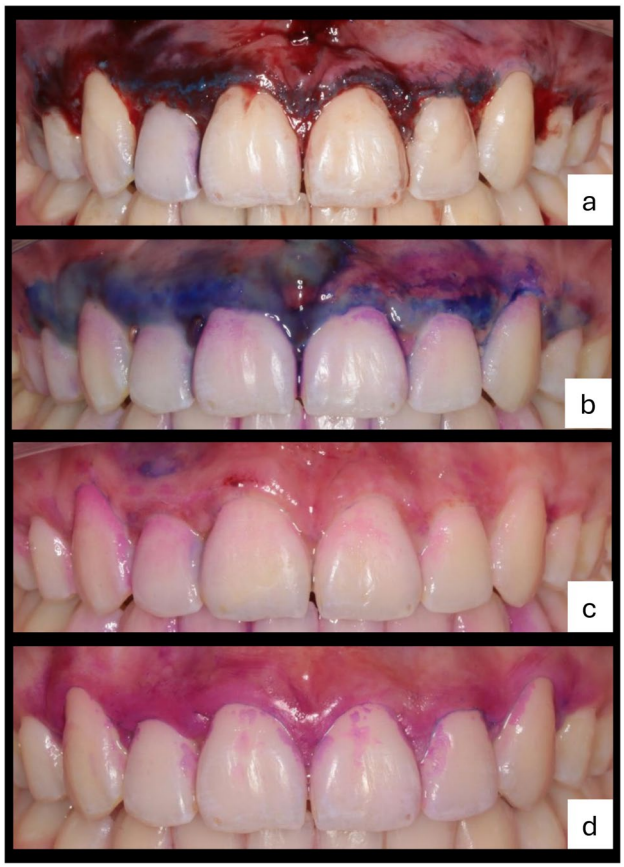


Fig. 2 Case example belonging to the LLLT group. **a**. Baseline **b**. Day 3 **c**. Day 7 **d**. Day 14

Image calibration was performed by measuring the clinical crown length of the maxillary right central incisor (from incisal edge to gingival margin) intraorally and comparing it to its corresponding dimension in the photograph to establish a pixel-to-millimeter ratio. Using the calibrated images, the boundaries of the surgical wound area were delineated, isolated, and transferred to a separate canvas. Images were converted to 32-bit format, and the stained (non-epithelialized) area was measured using standardized threshold settings applied uniformly across all images. The healing percentage at each time point was calculated by comparing the wound area at postoperative days 3, 7, and 14 to the baseline wound area using the following formula: Healing (%)=[(Baseline area−Area at time point)/Baseline area]×100.

Secondary outcomes: periodontal parameters

All clinical and digital measurements were performed by a single calibrated examiner who was blinded to the group allocations. All clinical periodontal parameters were recorded at six sites per tooth using a Williams periodontal probe at baseline and at 6 weeks postoperatively. Gingival inflammation was quantified using the GI in accordance with the

criteria originally established by Löe and Silness [28]. Gingival bleeding status was assessed dichotomously as present or absent at each probing site, and the FMBS was derived by calculating the proportion of bleeding-positive sites as a percentage of the total number of sites examined. PPD was measured from the gingival margin to the base of the sulcus or pocket at each site and recorded in millimeters [29].

Visual analogue scale (VAS)

Postoperative pain was assessed using a 10-cm visual analogue scale (VAS), a widely validated instrument for the quantification of acute pain intensity [30]. Patients were instructed to mark their perceived pain level on a horizontal line anchored by 'no pain' (0) at one end and 'worst imaginable pain' (10) at the other. Scores were categorized as follows: 0=no pain, 1–3=mild pain, 3.1–6=moderate pain, and 6.1–10=severe pain. VAS assessments were recorded on postoperative days 1, 3, 7, and 14.

Statistical analysis

Descriptive statistics were presented using frequency, percentage, mean, standard deviation, median, minimum, maximum, and the 25th–75th percentile (Q1–Q3) values. The assumption of normality was assessed using the Shapiro–Wilk test, along with visual methods including histograms and Q-Q plots, as well as skewness and kurtosis values. For comparison of numerical data between two independent groups, the Mann–Whitney U test was used when the data did not follow a normal distribution. In the analysis of two related measurements, the Wilcoxon Signed-Rank Test was applied if the distribution deviated from normality. For more than two related measurements, when the data in both groups met the assumption of normality, a Mixed Design ANOVA (Mixed-ANOVA) was employed. The assumption of sphericity was assessed using Mauchly's test of sphericity. When the assumption was violated and the Greenhouse–Geisser epsilon (ε) value was less than 0.75, Greenhouse–Geisser-adjusted p-values were reported. For categorical variables, Pearson's Chi-Square Test was used when less than 20% of the cells had expected counts below 5. A p-value less than 0.05 was considered statistically significant. All analyses were performed using the SPSS version 23.0 (IBM Corp., Armonk, NY, USA) statistical software package.

Results

Demographic characteristics and study flow

A total of 64 participants were enrolled and randomized. All participants completed the assigned interventions, and no adverse events or side effects were observed throughout the study period. Participants were equally allocated to the LLLT group ($n=32$) and the HA spray group ($n=32$). The overall sample comprised 21 males (32.8%) and 43 females (67.2%). The mean age of the participants was 18.41 ± 3.58 years, ranging from 13 to 28 years (median = 17). No statistically significant difference was observed in gender distribution between the groups, although the difference approached statistical significance (χ^2 [1] = 3.70, $p=0.062$). In the LLLT group, 43.8% were male and 56.3% were female, whereas in the HA spray group, 21.9% were male and 78.1% were female. No statistically significant difference was found between the groups in terms of mean age (U = 389, $p=0.097$). The mean age was 19.09 ± 3.47 years in the LLLT group and 17.72 ± 3.62 years in the HA spray group (Table 1).

Table 1 Gender and age distribution in the two study groups

Variable	LLLT ($n=32$)	HA spray ($n=32$)	Test statistic	p
Gender, n (%)			$\chi^2(1)=3.70$	0.062[a]
Male	14 (43.8)	7 (21.9)		
Female	18 (56.3)	25 (78.1)		
Age (years)			U = 389.0	0.097[b]
Mean ± SD	19.09 ± 3.47	17.72 ± 3.62		
Median (Q1–Q3)	20.00 (16.50–21.50)	17.00 (15.00–20.50)		

Data are presented as n (%) for categorical variables and as mean ± SD and median (Q1–Q3) for continuous variables

a Pearson's Chi-Square test; the reported test statistic is the Pearson χ^2 value with 1 degree of freedom

b Mann–Whitney U test; the reported test statistic is the Mann–Whitney U value

Statistical significance was set at $p<0.05$

LLLT low-level laser therapy; *HA* hyaluronic acid; *SD* standard deviation; *Q1–Q3*, 25th–75th percentiles; *U* Mann–Whitney U statistic; χ^2 Pearson chi-square statistic

Wound surface area measurements

At baseline, no statistically significant difference was observed between the two groups in wound surface area ($p>0.05$). The main effect of time was statistically significant ($p<0.001$, $\eta^2=0.907$), indicating that wound surface area decreased significantly over time irrespective of group allocation. The time × group interaction was not statistically significant ($p=0.068$, $\eta^2=0.048$), suggesting that the rate of wound area reduction over time did not differ significantly between the two groups. The main effect of group was statistically significant ($p=0.049$, $\eta^2=0.061$), indicating that, overall, the mean wound surface area was significantly lower in the HA spray group compared to the LLLT group (Table 2; Fig. 3).

Table 2 Wound surface measurements according to ImageJ analysis

	LLLT			HA SPRAY					
	n	Mean ± SD	Median (Q1-Q3)	*n*	Mean ± SD	Median (Q1-Q3)	Effect Type	*p*	ES
Baseline	32	183.08 ± 54.3	196.16(150.01–208.55)	32	178.76 ± 48.13	181.3(150.98–209.82)	Main effect	**< 0.001**	0.907
Day 3	32	108.08 ± 43.6	109.67(81.92–128.4)	32	81.39 ± 25.9	86.27(63.38–95.14)	Interaction effect	**0.068**	0.048
Day 7	32	50.55 ± 21.27	53.2(33.93–65.93)	32	35.56 ± 12.29	37.19(25.15–42.88)	Group effect	**0.049**	0.061
Day 14	32	22.74 ± 11.25	21.69(14.48–27.41)	32	16.73 ± 5.9	16.48(11.41–21.72)			

Since the sphericity assumption was not met and ε < 0.75. the Greenhouse–Geisser correction was applied for the interpretation of the Mixed ANOVA results. *ES* Effect Size. Partial Eta Squared

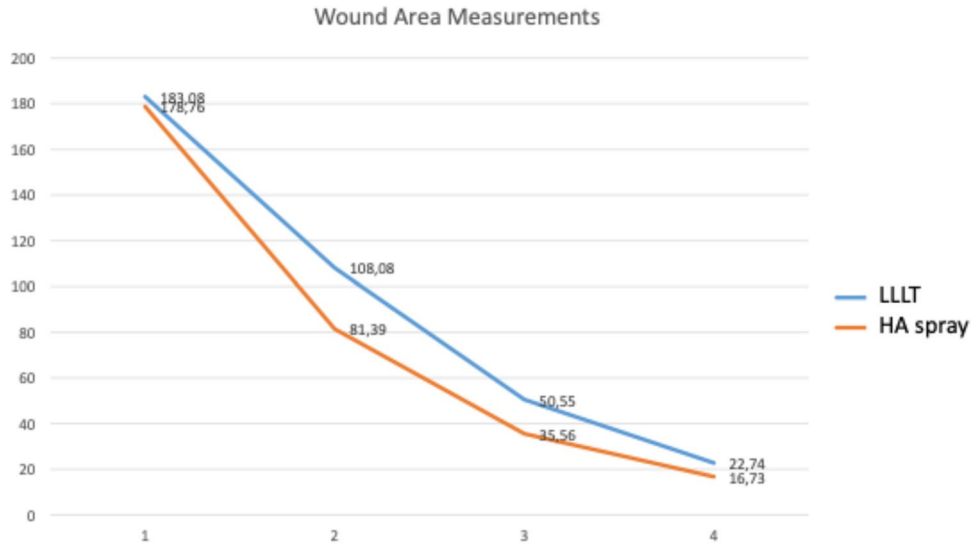


Fig. 3 Time-dependent changes in wound surface measurements according to ImageJ analysis

Healing percentages

In both groups, wound healing percentages demonstrated statistically significant increases across postoperative days 3, 7, and 14 ($p < 0.001$ for all pairwise time-point comparisons), confirming a progressive healing trajectory in both treatment arms. In intergroup comparisons, the HA spray group demonstrated significantly higher healing percentages than the LLLT group on day 3 (U = 157, $p < 0.001$), day 7 (U = 237, $p < 0.001$), and day 14 (U = 362, $p = 0.044$). The between-group mean differences favored HA spray by 12.33% on day 3 (95% CI: 7.89 to 16.77; Cohen's d = 1.39), 7.47% on day 7 (95% CI: 4.13 to 10.81; Cohen's d = 1.12), and 2.89% on day 14 (95% CI: 0.92 to 4.86; Cohen's d = 0.73). These findings indicate a large effect size in favor of HA spray during the early healing period, particularly on days 3 and 7, with a moderate-to-large effect persisting on day 14 (Table 3; Fig. 4).

Periodontal parameters

All periodontal parameters demonstrated statistically significant improvements from baseline to week 6 in both groups ($p < 0.001$ for all within-group comparisons). GI scores decreased significantly in both groups by week 6; however, no statistically significant intergroup difference was detected at baseline (U = 464, $p = 0.285$) or at week 6 (U = 512, $p = 0.999$) (Table 3). FMBS values followed a similar pattern, with significant within-group reductions by week 6 ($p < 0.001$) but no statistically significant between-group differences at either baseline (U = 470, $p = 0.569$) or week 6 (U = 497.5, $p = 0.843$) (Table 3). Likewise, PPD values decreased significantly in both groups by week 6 ($p < 0.001$), with no statistically significant intergroup differences at any time point ($p > 0.05$) (Table 3). Collectively, these findings indicate that both LLLT and HA spray were associated with comparable improvements in clinical periodontal parameters over the 6-week follow-up period.

Visual analogue scale (VAS) scores

VAS pain scores decreased significantly over time in both groups ($p < 0.001$ for all within-group time-point comparisons). On postoperative day 1, the HA spray group reported significantly lower pain scores compared to the LLLT group (U = 343, $p = 0.010$; mean difference: −0.41; 95% CI: −0.71 to −0.11; Cohen's d = −0.69), and this significant difference persisted on day 3 (U = 384, $p = 0.046$; mean difference: −0.25; 95% CI: −0.49 to −0.01; Cohen's d = −0.52), indicating small-to-moderate effect sizes in

Table 3 Between-group and within-group comparisons of wound healing percentage, clinical periodontal parameters and postoperative pain

	LLLT ($n=32$)			HA spray ($n=32$)			Between-group comparison	
	n	Mean ± SD	Median (Q1–Q3)	*n*	Mean ± SD	Median (Q1–Q3)	Test statistic (U)	*p*
Wound healing (%)								
Between-group comparisons								
Day 3	32	42.17 ± 10.58	44.60 (35.67–49.39)	32	54.50 ± 6.77	56.08 (49.19–59.76)	157.0	< 0.001‡
Day 7	32	72.71 ± 8.10	71.68 (67.04–79.18)	32	80.18 ± 4.87	79.51 (77.28–84.45)	237.0	< 0.001‡
Day 14	32	87.58 ± 4.91	89.85 (83.99–91.47)	32	90.47 ± 2.62	90.13 (88.63–92.50)	362.0	0.044‡
Within-group comparisons over time								
Overall change over time		$\chi^2(2)=64.00$	< 0.001§		$\chi^2(2)=64.00$	< 0.001§		
Day 3 vs. day 7		z = 4.94	< 0.001†		z = 4.94	< 0.001†		
Day 3 vs. day 14		z = 4.94	< 0.001†		z = 4.94	< 0.001†		
Day 7 vs. day 14		z = 4.94	< 0.001†		z = 4.94	< 0.001†		
Gingival Index (GI)								
Between-group comparisons								
Baseline	32	2.19 ± 0.40	2.00 (2.00–2.00)	32	2.09 ± 0.30	2.00 (2.00–2.00)	464.0	0.285
Week 6	32	0.44 ± 0.50	0.00 (0.00–1.00)	32	0.44 ± 0.50	0.00 (0.00–1.00)	512.0	0.999
Within-group comparison over time								
Baseline vs. week 6		z = 5.11	< 0.001†		z = 5.10	< 0.001†		
Full-Mouth Bleeding Score (FMBS, %)								
Between-group comparisons								
Baseline	32	19.97 ± 2.42	20.00 (18.00–21.50)	32	20.12 ± 2.17	20.00 (19.00–22.00)	470.0	0.569
Week 6	32	4.84 ± 1.82	5.00 (3.50–6.00)	32	4.91 ± 1.67	5.00 (4.00–6.00)	497.5	0.843
Within-group comparison over time								
Baseline vs. week 6		z = 4.94	< 0.001†		z = 4.95	< 0.001†		
Probing Pocket Depth (PPD, mm)								
Between-group comparisons								
Baseline	32	2.48 ± 0.36	2.50 (2.17–2.83)	32	2.46 ± 0.29	2.50 (2.17–2.67)	502.5	0.897
Week 6	32	1.23 ± 0.13	1.22 (1.15–1.30)	32	1.25 ± 0.12	1.23 (1.20–1.33)	485.5	0.718
Within-group comparison over time								
Baseline vs. week 6		z = 4.94	< 0.001†		z = 4.94	< 0.001†		
Visual Analogue Scale (VAS, 0–10 cm)								
Between-group comparisons								
Day 1	32	4.41 ± 0.56	4.00 (4.00–5.00)	32	4.00 ± 0.62	4.00 (4.00–4.00)	343.0	0.010‡
Day 3	32	2.56 ± 0.50	3.00 (2.00–3.00)	32	2.31 ± 0.47	2.00 (2.00–3.00)	384.0	0.046‡
Day 7	32	0.25 ± 0.44	0.00 (0.00–0.50)	32	0.19 ± 0.40	0.00 (0.00–0.00)	480.0	0.549
Within-group comparisons over time								
Overall change over time		$\chi^2(2)=63.51$	< 0.001§		$\chi^2(2)=63.05$	< 0.001§		
Day 1 vs. day 3		z = 4.99	< 0.001†		z = 4.93	< 0.001†		
Day 1 vs. day 7		z = 5.01	< 0.001†		z = 5.05	< 0.001†		
Day 3 vs. day 7		z = 5.04	< 0.001†		z = 5.06	< 0.001†		

Data are presented as mean ± SD and median (Q1–Q3). Wound healing percentage denotes the percentage reduction from baseline in the non-epithelialized, disclosing solution stained wound surface area measured using ImageJ planimetry.Between-group comparisons at each time point were performed using the Mann–Whitney U test; the reported test statistic is the Mann–Whitney U value.For variables measured at three time points (wound healing percentage and VAS), the overall within-group change over time was evaluated using the Friedman test, with the χ^2 statistic and 2 degrees of freedom reported. This was followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction for three comparisons, using an adjusted significance level of $\alpha=0.017$. For variables measured at two time points (GI, FMBS, and PPD), the Wilcoxon signed-rank test constituted the overall within-group comparison. The reported test statistic for the Wilcoxon signed-rank test is the absolute standardized z value

‡ Statistically significant between-group difference at the corresponding time point (Mann–Whitney U test, $p<0.05$)

† Statistically significant within-group comparison based on the Wilcoxon signed-rank test. For wound healing percentage and VAS, † indicates a statistically significant pairwise comparison after Bonferroni correction (adjusted significance level $\alpha=0.017$). For GI, FMBS, and PPD, † indicates a statistically significant overall baseline-to-week 6 comparison ($p<0.05$)

§ Statistically significant overall within-group change over time (Friedman test, $p<0.05$)

LLLT low-level laser therapy; *HA* hyaluronic acid; *GI* gingival index; *FMBS* full-mouth bleeding score; *PPD* probing pocket depth; *VAS* visual analogue scale; *SD* standard deviation; *Q1–Q3* 25th–75th percentiles; *n* sample size; *U* Mann–Whitney U statistic; *z* standardized Wilcoxon signed-rank statistic; χ^2 Friedman chi-square statistic

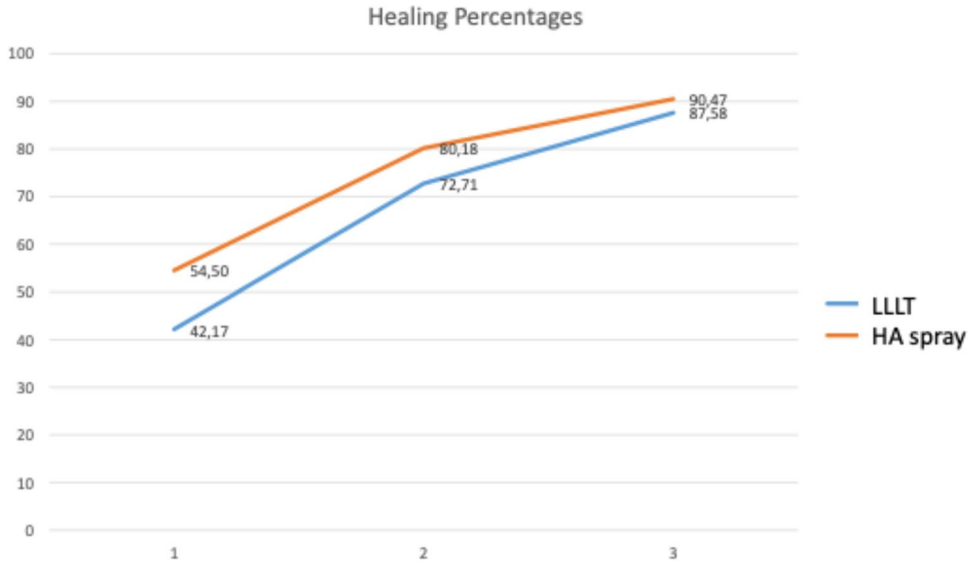


Fig. 4 Time-dependent changes in healing percentages

favor of HA spray for early postoperative pain control. By day 7, however, VAS scores were no longer significantly different between the groups (U = 480, $p = 0.549$; mean difference: −0.06; 95% CI: −0.27 to 0.15; Cohen's d = −0.14). VAS values on postoperative day 14 were not analyzed between groups, as most individuals in both cohorts reported a pain level of zero, creating a floor effect that hindered significant statistical comparison.

Discussion

Wound healing following surgical gingivectomy proceeds by secondary intention and is associated with a prolonged recovery period and increased patient morbidity compared with primary-intention healing [1]. This process involves inflammatory cell recruitment, cellular proliferation, extracellular matrix deposition, and progressive tissue remodeling [32, 33]. Given the clinical importance of accelerating early epithelialization and improving patient comfort, various adjunctive approaches have been investigated to enhance postoperative healing. While previous studies have demonstrated favorable effects of LLLT on wound healing and pain reduction following gingivectomy and other periodontal surgical procedures [7, 9, 22, 34], comparative data evaluating LLLT against other biologically active adjuncts remain limited. Accordingly, the present study was designed as a randomized controlled clinical trial to provide a head-to-head evaluation of LLLT and high-molecular-weight HA spray in postoperative wound healing and pain control following gingivectomy.

Since the study did not include a conventional surgical control group treated with gingivectomy alone, the findings should be interpreted as a comparison between two active adjunctive interventions rather than a demonstration of absolute efficacy relative to natural healing. Notably, to the best of our knowledge, this is the first randomized controlled trial to evaluate the effects of a high-molecular-weight HA spray formulation in the context of postoperative wound healing following gingivectomy.

Chronic inflammatory gingival overgrowth is frequently associated with orthodontic treatment, which promotes plaque retention through the presence of fixed appliances [9]. In the present study, 64 patients with orthodontic treatment-induced gingival overgrowth requiring gingivectomy were enrolled. Symmetrical anterior regions were specifically selected to facilitate standardized clinical assessment and enable reproducible ImageJ-based digital planimetric analysis. Although the sex distribution difference did not reach statistical significance ($\chi^2 = 3.70$, $p = 0.062$), the higher proportion of males in the LLLT group (43.8% vs. 21.9%) warrants consideration. Unlike dermal wounds, oral mucosal wound healing has been reported to favor males over females, a pattern attributed to differential sex hormone effects on the proliferative phase of tissue repair [35, 36]. If applicable to gingivectomy wounds, this baseline imbalance would be expected to favor the LLLT group; the superior early healing and analgesic outcomes observed in the HA spray group were therefore achieved despite this distributional asymmetry.

In both groups, healing percentages increased significantly over time. In the intergroup comparisons, the HA spray group demonstrated significantly higher healing percentages than the LLLT group on postoperative days 3, 7, and 14. These findings suggest that, while both modalities supported postoperative healing, HA spray may offer an additional advantage in enhancing early wound epithelialization after gingivectomy.

A recent systematic review [37] highlighted that the evidence regarding the efficacy of LLLT on wound healing remains heterogeneous, largely due to variations in laser parameters and application protocols; therefore, no universally accepted standardized LLLT protocol has yet been established for periodontal wound healing. In the present study, LLLT was performed using a 940 nm diode laser with a power density of 1.07 W/cm^2 on postoperative days 0, 3, 5, and 7. Each session lasted 120 s, delivering 360 J per session and an energy density of 128.4 J/cm^2, with a cumulative delivered energy of 1440 J over four sessions. Previous studies have reported favorable outcomes with diode lasers using different wavelengths, dosimetry parameters, and application schedules following gingivectomy and other periodontal surgical procedures [7, 11, 34, 38]. However, differences in treatment parameters should be considered when comparing the present findings with those of previous investigations. In a comparable study, George et al. [39] reported that both ozone therapy and LLLT significantly reduced wound surface area compared to the control group, although no significant difference was observed between the two active modalities. It is also noteworthy that postoperative analgesics were prescribed in many previous studies, which may have influenced the reported effects of LLLT on wound healing. Direct comparison between studies should be interpreted cautiously, as differences in patient characteristics, follow-up periods, adjunctive treatment protocols, postoperative analgesic regimens, and wound-healing assessment methods may substantially influence reported outcomes.

Over the past decade, HA has been increasingly incorporated into both non-surgical and surgical periodontal treatment protocols owing to its well-documented anti-inflammatory, analgesic, and tissue-regenerative properties [14]. Given its role as a key component of the gingival extracellular matrix, postoperative HA application may support tissue repair and wound healing [40]. In the present study, a 0.04% high-molecular-weight HA spray was used. Although HA has been administered in various formulations and concentrations for periodontal applications [1, 12, 13, 18, 41], no consensus currently exists regarding the optimal formulation or application protocol for periodontal wound management. When compared with previous gingivectomy studies, important differences in both LLLT and HA application protocols should be considered. Yakout et al. [1] used 980 nm PBMT at 0.2 W and applied 2% high-molecular-weight HA gel with a custom-made soft tissue guard appliance to prolong contact with the wound surface. Reddy et al. [11] applied 980 nm LLLT at 50 mW with an energy density of 4 J/cm^2 for three minutes and used 0.2% HA gel under periodontal dressing. In contrast, the present study used a 940 nm diode laser protocol and a 0.04% high-molecular-weight HA spray applied twice daily for seven days without a tissue guard or periodontal dressing. Therefore, differences in laser wavelength, power output, energy density, application schedule, HA concentration, formulation, and wound contact time may partly explain the differences observed between the present findings and previous reports.

The higher early healing percentages observed in the HA spray group may be partially associated with both the biological and physical properties of hyaluronic acid. In addition to its anti-inflammatory and tissue-supportive effects, HA spray may provide a transient coating over the wound surface. This possible barrier-like effect may help reduce external irritation and create local conditions that are more favorable for epithelial repair. However, these mechanistic interpretations should be considered cautiously, as the present study did not include histological, immunohistochemical, or biomolecular analyses. Moreover, postoperative healing is a multifactorial outcome that may also be influenced by wound characteristics, oral hygiene status, patient compliance, and individual healing capacity.

VAS pain scores decreased significantly over time in both treatment groups. The HA spray group reported significantly lower pain scores than the LLLT group on postoperative days 1 and 3, whereas no significant intergroup difference was detected by day 7. These early differences may be clinically relevant, as postoperative discomfort during the initial days following periodontal surgery can influence patient comfort, treatment compliance, and overall perception of therapeutic success. The lower early pain scores observed in the HA group may be partially associated with the anti-inflammatory and tissue-supportive properties of HA, as well as its potential to provide a transient coating over the wound surface. This possible protective effect may help reduce mechanical irritation and exposure to oral stimuli during the early postoperative phase. However, patient-reported pain is a multifactorial outcome and may be influenced by analgesic use, wound size and characteristics, individual pain threshold, oral hygiene status, and patient compliance. Therefore, the mechanisms underlying the observed differences in pain scores should be interpreted with caution. Similar findings have been reported in previous studies suggesting that HA application, photobiomodulation, and ozone therapy may contribute to reduced postoperative pain, VAS scores, and inflammation after gingival surgery [1, 13, 18, 26, 39]. With regard to clinical periodontal parameters, both interventions yielded comparable outcomes. Significant reductions in GI, FMBS, and PPD values were observed in both groups over the 6-week follow-up period, with no statistically significant intergroup differences at any time point. These findings are consistent with those reported by Reddy et al. [11], who similarly demonstrated statistically significant improvements in gingival index, plaque index, and gingival

enlargement index following gingivectomy, irrespective of the adjunctive modality employed. Importantly, these periodontal parameters reflect longer-term clinical tissue stabilization and plaque-related inflammatory changes, rather than early wound surface epithelialization alone. Therefore, although HA spray was associated with higher early epithelialization percentages and lower pain scores during the initial postoperative period, both HA spray and LLLT resulted in comparable improvements in periodontal clinical parameters over the 6-week follow-up period. This distinction suggests that the potential clinical advantage of HA spray was mainly observed during the early postoperative healing phase, whereas longer-term periodontal outcomes were similar between the two interventions. For the assessment of wound epithelialization, Mira-2-Ton erythrosine-based disclosing solution was used as an alternative to hydrogen peroxide, which has been reported to exert cytotoxic effects on human gingival fibroblasts and may potentially compromise the healing process [41]. The degree of epithelialization was clinically evaluated by an experienced and calibrated periodontist using standardized digital planimetric analysis. Although this method provided a practical and reproducible approach for quantifying wound healing, future studies incorporating histological assessment or biomolecular analyses may provide more detailed insights into the underlying tissue repair mechanisms.

Study limitations

Several limitations of the present study warrant acknowledgment. First, the absence of a conventional gingivectomy-only control group inherently limits the interpretation of the absolute efficacy of either adjunctive treatment; accordingly, the findings should be regarded as a direct comparison between two active interventions rather than as evidence of superiority over natural wound healing. Second, the relatively short follow-up period predominantly captures early postoperative healing dynamics; therefore, longer-term longitudinal studies are warranted to determine whether the early advantages observed with HA spray translate into sustained clinical benefits. Third, the study population comprised predominantly young, systemically healthy individuals who underwent gingivectomy following orthodontic treatment; consequently, the generalizability of these findings to older patients, smokers, or individuals with systemic conditions that may impair wound healing remains uncertain. Fourth, although digital planimetric analysis provided a practical and reproducible assessment methodology, histological evaluation of soft tissue repair would offer more objective evidence regarding the cellular and molecular mechanisms underlying the observed healing differences. Fifth, all participants received the same medication regimen of 400 mg of ibuprofen twice daily for a duration of 5 days; however, the actual medication usage was not documented, which may represent a potential confounding variable in relation to VAS pain scores. Sixth, VAS pain scores should be interpreted with caution, as the open-label design precluded participant blinding, introducing a risk of performance and expectancy bias for patient-reported outcomes.

Finally, future multicenter studies with larger sample sizes are recommended to further elucidate the comparative effects of LLLT and HA spray on wound healing across diverse periodontal surgical procedures and patient populations.

Conclusion

Within the limitations of this randomized controlled clinical trial, both 940 nm diode laser therapy and 0.04% high-molecular-weight HA spray were well tolerated as adjunctive modalities following gingivectomy, with no adverse events observed during the study period. HA spray was associated with greater early wound healing and lower early postoperative pain scores compared with LLLT, whereas both interventions resulted in comparable improvements in periodontal clinical parameters during follow-up. These findings suggest that topical HA spray may represent a practical and patient-friendly adjunctive option for supporting early postoperative recovery following gingivectomy. Further multicenter studies with longer follow-up periods, histological or biomolecular assessments, and more diverse patient populations are warranted to confirm these findings and clarify the underlying healing mechanisms.

Author contribution Nezahat Arzu Kayar: Pproject administration, conceptualization, methodology, supervision, writing—original draft, writing—review and editing. Ipek Ozgu: Data curation, visualization, writing—original draft, Fatma Koksel Babun: Investigation, data curation, resources. Aysenur Cohadar Kaya: Investigation, methodology. Deniz Ozel: Formal analysis, software, validation. Esra Bolat Gümüs:: Conceptualization, Validation Mükerrem Hatipoğlu: : Conceptualization, supervision. All approved the final version for publication.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval This study was conducted upon receipt of ethical approval from the Akdeniz University Faculty of Medicine Ethics Committee (KAEK-467) and was performed in accordance with the 1964 Declaration of Helsinki and its later amendments. All participants were fully informed about the study objectives and procedures and provided written informed consent prior to enrollment. For participants under 18 years of age, written informed consent was obtained from a parent or legal guardian, and written assent was obtained from the minor participant, in accordance with the requirements of the ethics committee.This study was conducted upon receipt of ethical approval from the Akdeniz University Faculty of Medicine Ethics Committee (KAEK-467) and was performed in accordance with the 1964 Declaration of Helsinki and its later amendments. All participants were fully informed about

the study objectives and procedures and provided written informed consent prior to enrollment. For participants under 18 years of age, written informed consent was obtained from a parent or legal guardian, and written assent was obtained from the minor participant, in accordance with the requirements of the ethics committee.

Consent to participate All enrolled patients signed an informed consent before their participation.

Consent to publish Patients signed an informed consent regarding publishing their data and photographs.

Clinical trial registration The study was registered in ClinicalTrials.gov (NCT07034625).

Competing interests The authors declare no competing interests.

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